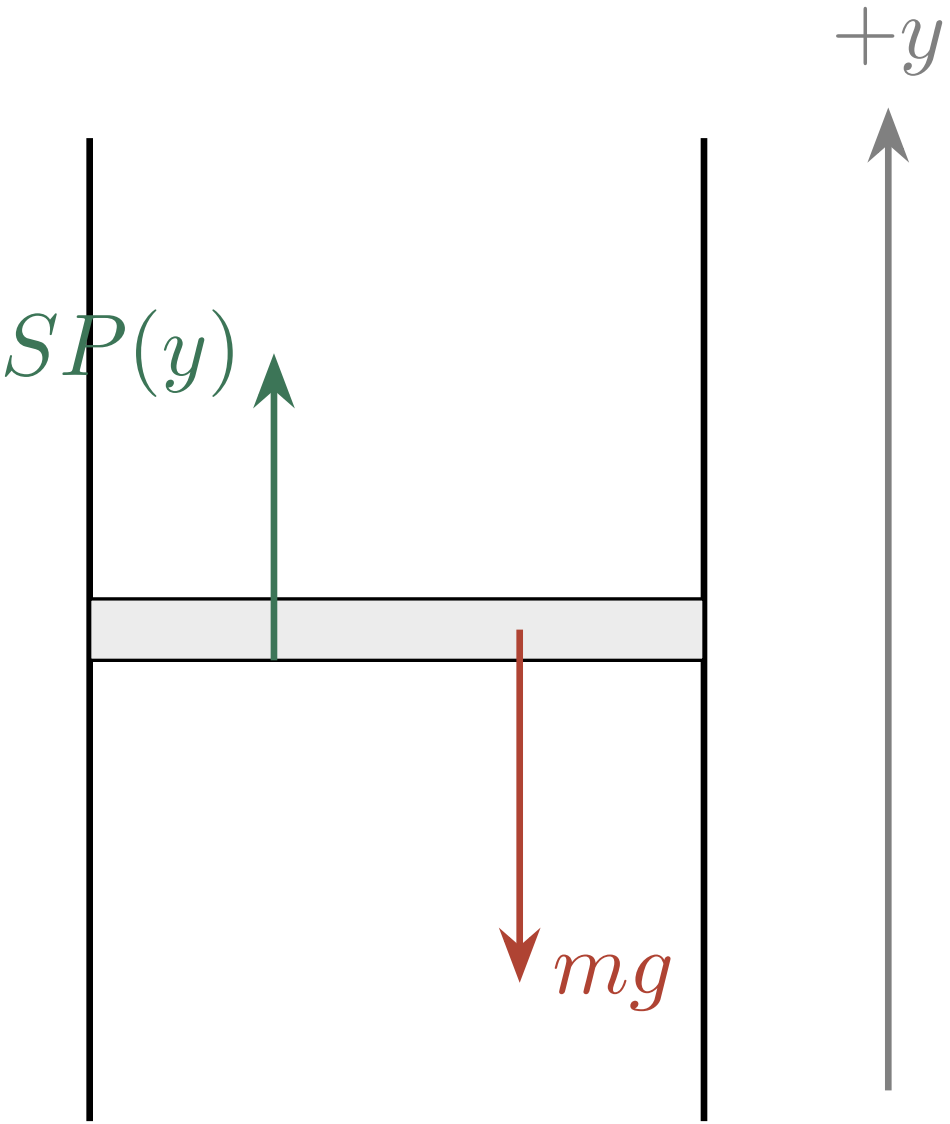
$$m\ddot{y} = SP(y) - mg$$



下方气体体积 $V_0 + Sy$